



AMD DEVELOPER HACKATHON

GEOCLUSTER 2.0

AI Geospatial Assistant

Satellite imagery retrieval, GIS processing, and a natural-language AI assistant — in one desktop application.

Built on Python · PyQt5 · Sentinel-2 · GPT-OSS-20B · AMD Developer Cloud

THE PROBLEM

GIS software was built for specialists, not for the people who need it fastest.



Steep learning curve

Menus, coordinate systems, and file formats stand between a question and an answer.



Manual data hunting

Finding and downloading the right satellite scene is its own multi-step chore.



Disconnected tools

Fetching imagery and processing it usually means switching between separate apps.



*Too many tools.
Too many steps.*

THE SOLUTION

Ask for what you need. GEOCLUSTER does the rest.

GEOCLUSTER 2.0 wraps satellite retrieval and GIS processing behind a single natural-language assistant — so a plain English sentence replaces a whole workflow.



"Show me NUST imagery and cluster it into 5 zones."



Fetching Sentinel-2 · running K-Means · rendering result

1

NATURAL-LANGUAGE INTERFACE

8

GIS OPERATIONS UNIFIED

0

MANUAL DOWNLOADS NEEDED

KEY FEATURES

Everything a geospatial workflow needs, unified behind one assistant.



AI Assistant

GPT-OSS-20B answers GIS & remote sensing questions directly.



Sentinel-2 Retrieval

Pulls real satellite imagery for any named location, on demand.



Natural Language Commands

Plain English replaces menus — "cluster this into 5 zones."



Image Processing

Brightness, threshold, negative, and mean filtering, all built in.



K-Means Clustering

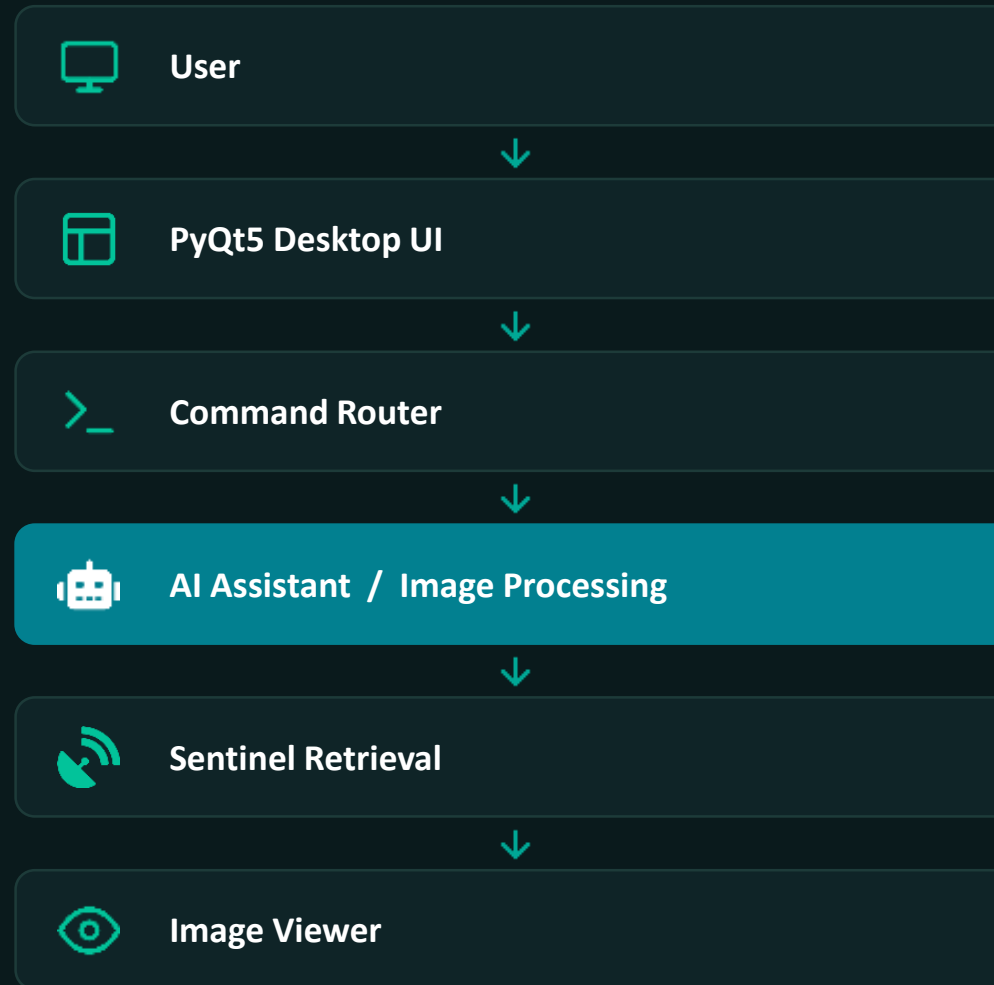
GPU-ready unsupervised classification of imagery.



Measurement Tools

Distance and histogram analysis on any loaded image.

From a typed sentence to a rendered result.



Design principle

One router decides whether a prompt needs the AI assistant, an image operation, or a satellite fetch — then hands off to the right module.

TECHNOLOGY STACK

Built on a lean, production-ready stack.



Python



PyQt5



OpenCV



NumPy



Matplotlib



Pillow



GPT-OSS-20B



Sentinel-2



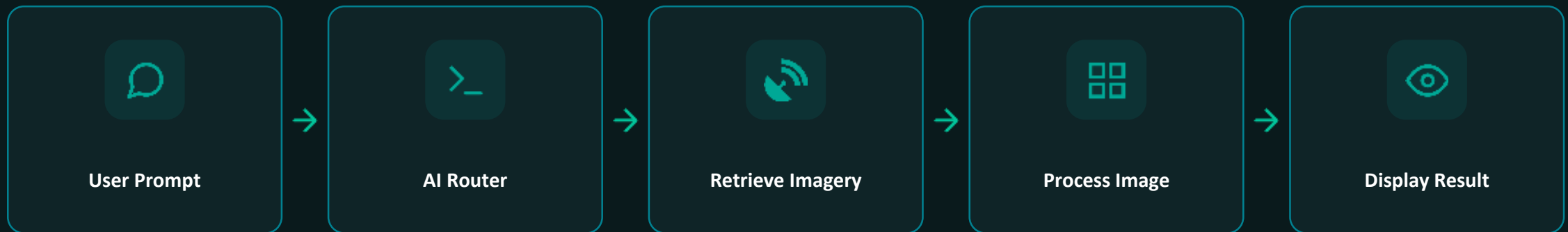
Docker



AMD Dev Cloud

DEMO WORKFLOW

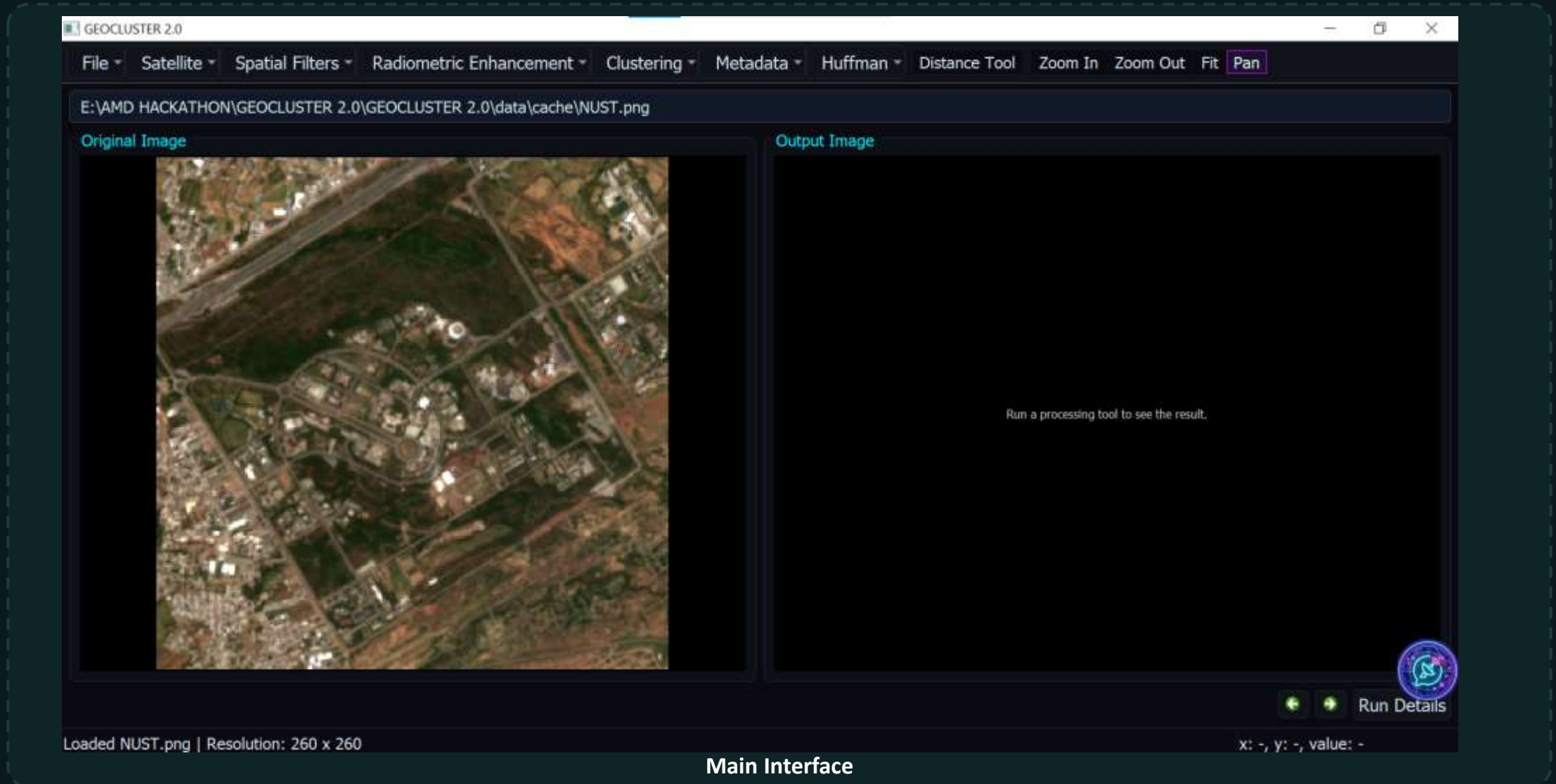
One sentence, five automated steps.



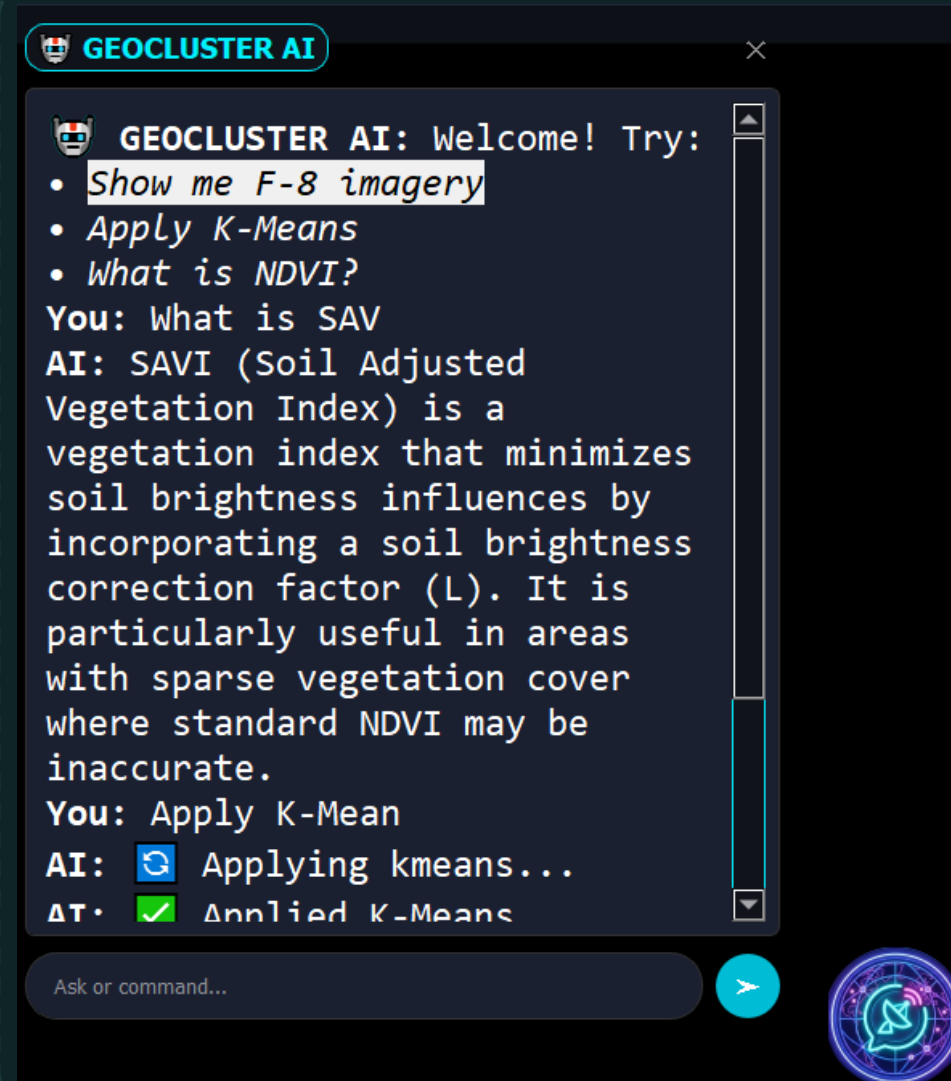
Example: "Show me NUST imagery and cluster it into 5 zones."

LIVE DEMO

See it in action.



See it in action.



AI Assistant Panel



FUTURE WORK

Where GEOCLUSTER goes next.



NDVI / SAVI

Built-in vegetation index calculations.



Land Cover Classification

Automated supervised classification workflows.



Multi-Satellite Support

Landsat and other sources alongside Sentinel-2.



Vector GIS Tools

Shapefile and vector layer support.



Cloud Deployment

Run GEOCLUSTER without a local install.



Web Version

Browser-based access for broader reach.



Thank You

GEOCLUSTER 2.0 — AI Geospatial Assistant

AMD Developer Hackathon · Built by Team Quantum_Coders



<https://github.com/Furqan6701/GEOCLUSTER-2.0>